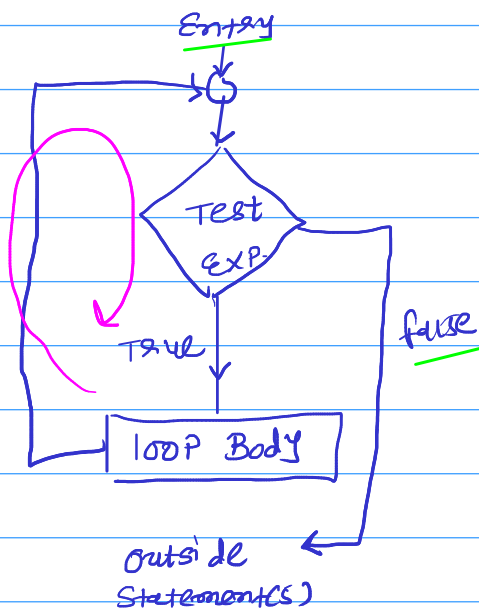
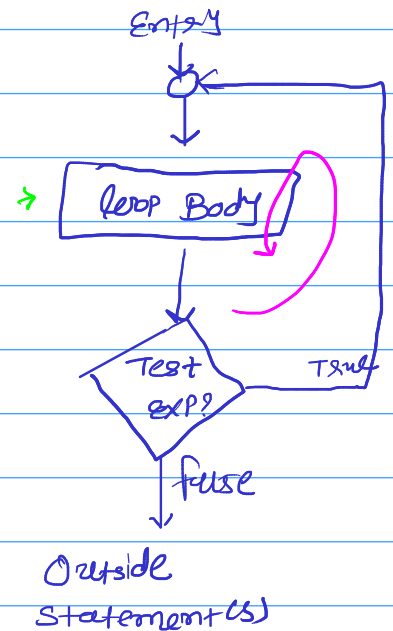


Loops :-

① Entry Control for, while



② Exit Control do-while



→ Sentinel Controlled loops

V/s

Counter Controlled loops

```

int i = 1;
for (; i != 5; i++) {
    scanf("%d", &i);
    printf("%d", i);
}
  
```

o/p :- ?

```

int i = 1;
for (; i != 5; i++) {
    printf("%d", i);
}
  
```

o/p :- 1234

(*)

for loop $\Rightarrow$

for (^① initialization ; ^② Condition ; ^④ inc/dec) {
 // Body of for loop ^③
}
// Outside statements

1 2 3 4 2 3 4 2 3 4 2 3 4 2 3 4 2

```
#include<stdio.h>
int main() {
```

```
    int no1=5;
```

```
    /*
    for(; no1 > 0 ; no1 --){
    }
    */
```

```
    for(; no1 > 0 ; no1 --);
        printf("Inside\n");
        printf("\nOutside");
```

```
    return 0;
}
```

Output :-

Inside
Outside

```
#include<stdio.h>
```

```
int main() {
```

```
    int no1=5;
```

```
    /*
```

```
    for(; no1 > 0 ; no1 --){
```

```
    }
```

```
    */
```

```
    for(; no1 > 0 ; no1 --);
```

```
    printf("\nInside: %d",no1);
```

```
    printf("\nOutside : %d",no1);
```

```
    return 0;
```

```
}
```

O/P:-

Inside : 0
Outside : 0

```
#include<stdio.h>
```

```
int main() {
```

```
    int no1=5;
```

```
    /*
```

```
    for(; no1 > 0 ; no1 --){
```

```
    }
```

```
    */
```

```
    for(; no1 > 0 ; no1 --)
```

```
        printf("\nInside: %d",no1);
```

```
    printf("\nOutside : %d",no1);
```

```
    return 0;
```

```
}
```

Inside: 5
Inside: 4
Inside: 3
Inside: 2
Inside: 1
Outside : 0



while loop :-

```
#include<stdio.h>
int main() {
```

```
    int no1=5;
```

Output :-

```
    /*
    while(condition){
        Statements
    }
    */
```

5

4

3

2

1

```
    while(no1>0){
        printf("\n%d",no1);
        no1--;
    }
```

```
}
```

```
#include<stdio.h>
int main() {
```

```
    int no1=5;
```

```
    /*
    while(condition){
        Statements
    }
    */
```

5

4

3

2

1

0

```
    while(no1>0){
        printf("\n%d",no1);
        --no1;
    }
    printf("\n%d",no1);
```

```
}
```

```
#include<stdio.h>
int main() {
```

```
    int no1=5;
```

```
    /*
```

```
    while(condition){
        Statements
```

```
    }
```

```
    */
```

```
    while(no1>0){
        printf("\n%d",--no1);
```

```
    }
```

```
    printf("\n%d",no1);
```

```
}
```

4

3

2

1

0

0

⑧ *

do - while

```
#include<stdio.h>
int main() {

    int no1=5;

    /*
        Statements
    */while(condition);

    /*
        do{
            no1--;
            printf("%d",no1);
        }while(no1>0);

        //printf("\n%d",no1);
        return 0;
    }
```

43210

```
#include<stdio.h>
int main() {
```

```
    int no1=5;
```

```
    /*
```

```
    do{
        Statements
    }while(condition);
```

```
    /*
```

```
    do{
        printf("%d",no1);
        no1--;
    }while(no1>0);
```

```
    //printf("\n%d",no1);
```

```
}
```

54321

```
#include<stdio.h>
```

```
int main() {
```

```
    int no1=5,no2=5;
```

```
    printf("\n Output by do-while loop\n");
```

```
    do{
```

```
        printf("%d",no1);
```

```
        no1--;
```

```
    }while(no1>0);
```

Output by do-while loop

54321

Output by while loop

54321

```
    printf("\n Output by while loop\n");
```

```
    while(no2>0)
```

```
    {
```

```
        printf("%d",no2);
```

```
        no2--;
```

```
    }
```

```
    //printf("\n%d",no1);
```

```
    return 0;
```

```
}
```

```
#include<stdio.h>
```

```
int main() {
```

```
    int no1=0,no2=0;
```

```
    printf("\n Output by do-while loop\n");
```

```
    do{
```

```
        printf("%d",no1);
```

```
        no1--;
```

```
    }while(no1>0);
```

Output by do-while loop

0

Output by while loop

```
    printf("\n Output by while loop\n");
```

```
    while(no2>0)
```

```
    {
```

```
        printf("%d",no2);
```

```
        no2--;
```

```
    }
```

```
    //printf("\n%d",no1);
```

```
    return 0;
```

```
}
```

```
#include<stdio.h>
int main() {
```

```
    int no1=5;
```

```
    do
```

```
        printf("%d",no1--);
```

```
        printf("Hello");
```

```
    while(no1>0);
```

```
    return 0;
```

```
}
```

error: expected 'while' before 'printf'
printf("Hello");
.